Cambridge (CIE) A Level Computer Science

Your notes

Data Integrity

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Validation & verification

What is validation?

- Validation is an **automated process** where **a computer checks** if a user input is **sensible** and meets the program's requirements
- There are **seven categories** of validation which can be carried out on **fields** and **data types,** these are
  - **Range check**
  - **Format check**
  - **Length check**
  - **Presence check**
  - **Existence check**
  - **Limit check**
  - **Check digit**
- There can be occasions where **more than one type of validation will be used** on a field
- An example of this could be a password field which could have a **length**, **presence** and **type** check on it

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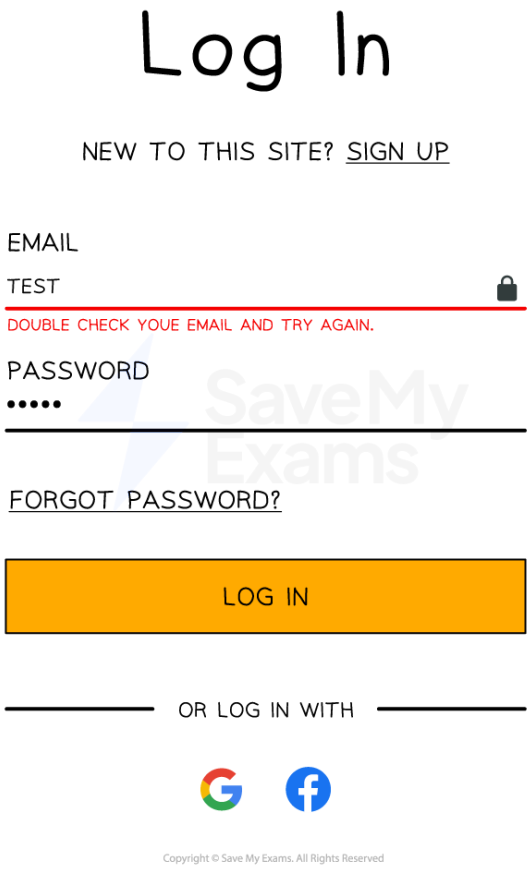

What is verification?

- Verification is the **act of checking data is accurate** when being **transferred** or **entered** into a system
- Verification methods include:
  - **Parity check (transfer)**
  - **Checksum (transfer)**
  - **Double entry checking (entry)**
  - **Visual checks (entry)**

Validation methods

Validation type	Definition	Example
Range check	Ensures a number falls within a set range	Validating that a percentage is between 0 and 100

Limit check	Checks that a value does not exceed a maximum limit	Ensuring no more than 5 items can be bought in a special offer
Length check	Checks the length of a string	Confirming a PIN is exactly 4 digits long
Type check	Checks that the input is of the correct data type	Ensuring age is entered as a whole number
Presence check	Checks that data has been entered and the field is not blank	Making sure a registration form is not submitted with blank fields
Existence check	Checks that a referenced value exists in a database or list	Confirming a student ID entered exists in the school records
Format check	Ensures data is in the correct format (e.g. patterns)	Making sure an email address contains '@' and a domain like '.com'

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What is a check digit?

- A check digit is the **last digit** included in a code or sequence, used to **detect errors** in **numeric data entry**
- Examples of errors that a check digit can help to identify are:
  - **Incorrect digits entered**
  - **Omitted or extra digits**
  - **Phonetic errors**
- Added to the **end of a numerical sequence** they ensure **validity** of the data
- Calculated using **standardised algorithms** to ensure widespread **compatibility**
- Examples of where check digits can be used include:
  - **ISBN book numbers**
  - **Barcodes**

ISBN book numbers

- Each book has a **unique** ISBN number that **identifies** the book
- A standard ISBN number may be ten digits, for example, 965–448–765–9
- The check digit value is the **final digit** (9 in this example).
- This number is chosen specifically so that when the algorithm is completed the result is a whole number (an integer) with no remainder parts

- A check digit algorithm is performed on the ISBN number and **if the result is a whole number,** then the **ISBN is valid**

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Barcodes

- Barcodes consist of **black and white lines** which can be scanned using barcode scanners
- Barcode scanners **shine a laser** on the black and white lines which **reflect light** into the scanner
- The scanner reads the **distance between these lines as numbers** and can identify the item
- Barcodes also use a **set of digits** to **uniquely identify each item**
- The **final digit** on a barcode is usually the **check digit**, this can be used to **validate** and **authenticate** an item

Verification methods

What is a parity check?

- A parity check determines whether **bits in a transmission** have been **corrupted**
- Every byte transmitted has **one of its bits** allocated as a **parity bit**
- The sender and receiver must **agree** before transmission whether they are using **odd** or **even parity**
- If **odd parity** is used then there must be an **odd number of 1's in the byte**, including the parity bit
- If **even parity** is used then there must be an **even number of 1's in the byte**, including the parity bit
- The **value of the parity bit** is determined by **counting the number of 1's in the byte**, including the parity bit
- If the number of 1's **does not match the agreed parity** then an **error has occurred**
- Parity checks **only check that an error has occurred**, they do not reveal where the error(s) occurred

Even parity

- Below is an arbitrary binary string

EVEN Parity bit	Byte						
0	1	0	1	1	0	1	0

- If an **even parity bit** is used then **all bits** in the byte, including the parity bit, must **add up to an even number**

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- There are four 1's in the byte
- This means the parity bit must be 0 otherwise the whole byte, including the parity bit, would add up to five which is an odd number

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Odd parity

- Below is an arbitrary binary string

ODD Parity bit	Byte						
1	1	0	1	1	0	1	0

- If an **odd parity bit** is used then **all bits** in the byte, including the parity bit, must **add up to an odd number**
  - There are four 1's in the byte. This means the parity bit must be a 1 otherwise the whole byte, including the parity bit, would add up to four which is an even number
- The table below shows a number of examples of the agreed parity between a sender and receiver and the parity bit used for each byte

Example #	Agreed parity	Parity bit	Main bit string							Total number of 1's
#1	ODD	0	1	1	0	1	0	1	1	5
#2	EVEN	1	0	0	0	1	0	0	0	2
#3	EVEN	1	0	1	0	1	1	1	1	6
#4	ODD	1	0	1	1	1	0	0	1	5
#5	ODD	1	1	0	1	0	1	0	1	5
#6	EVEN	0	1	0	0	1	1	1	0	4

- Example #1: The agreed parity is **odd**. All of the 1's in the main bit string are added **(5)**. As this number is **odd already** the **parity bit** is set to **0** so the whole byte **stays odd**
- Example #2: The agreed parity is **even**. All of the 1's in the main bit string are added **(1)**. As this number is **odd** the **parity bit** is set to **1** to make the **total number of 1's even (2)**
- Example #6: The agreed parity is **even**. All of the 1's in the main bit string are added **(4)**. As this number is **even already** the **parity bit** is set to **0** so the whole byte **stays even**

How do errors occur?

- When using parity bits, an **error** occurs when the number of **total bits does not match the agreed parity**

- Bits can be **flipped** or **changed** due to **interference** on a wire or wirelessly due to **weather** or **other signals**

Example #	Agreed parity	Parity bit	Main bit string							Total number of 1's	Error
#1	ODD	1	1	1	0	1	0	1	1	6	Error
#2	EVEN	1	0	0	0	1	0	0	0	2	No error
#3	EVEN	1	0	1	1	1	1	1	1	7	Error
#4	ODD	1	0	1	1	1	0	0	1	5	No error
#5	ODD	1	1	0	1	0	1	1	1	6	Error
#6	EVEN	0	1	0	0	0	1	1	0	3	Error

- Parity checks are **quick** and **easy to implement** but **fail to detect bit swaps** that cause the parity to remain the same

What are parity byte & block checks?

- Parity blocks and parity bytes can be used to **check an error has occurred** and **where the error is located**
- Parity checks on their own **do not pinpoint where errors in data exist**, only that an error has occurred
- A parity block consists of **a block of data** with the number of 1's totalled **horizontally** and **vertically**
- A parity byte is also sent with the data which contains the **parity bits** from the **vertical parity calculation**
- Below is a parity block with a parity byte at the bottom and a parity bit column in the second column

ODD	Parity bit	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
Byte 1	0	1	1	0	1	0	1	1
Byte 2	0	0	0	0	1	0	0	0
Byte 3	1	0	1	0	1	1	1	1

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Byte 4	1	0	1	1	1	0	0	1
Byte 5	1	1	0	1	0	1	0	1
Byte 6	1	1	0	0	1	1	1	0
Byte 7	0	0	1	1	1	1	1	0
Byte 8	0	1	0	1	1	0	0	0
Parity byte	0	1	1	1	1	1	1	1

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- The above table uses **odd parity**
- Each **byte row** calculates the **horizontal parity** as a parity bit **as normal**
- Each **bit column** calculates the **vertical parity** for each row, the **parity byte**
- It is **calculated before transmission** and **sent with the parity block**
- Each **parity bit tracks if a flip error** occurred in a byte while the **parity byte** calculates if an **error occurred in a bit column**
- By **cross referencing** both **horizontal and vertical** parity values the **error can be pinpointed**
- In the above example the **byte 3 / bit 5 cell** is the error and **should be a 0 instead**
- The error could be **fixed automatically or a retransmission request could be sent** to the sender

What is a checksum?

- A checksum is a value that can be used to determine if data has been **corrupted or altered**
- It indicates whether data **differs from its original form** but **does not** specify where
- Checksums are **calculated using an algorithm** and the value is **added to the transmission**
- The receiving device **re-calculates** the checksum and **compares** to the original
- If the **checksums do not match**, it is assumed an error has occurred

What is double entry checking?

- Double entry checking involves **entering the data twice** in separate input boxes and then comparing the data to ensure they both match
- If they do not, an error message is shown

What is a visual check?

- A visual check involves the user **visually checking the data on the screen**
- A popup or message then **asks if the data is correct** before proceeding
- If it isn't the **user then enters the data again**

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